Arrangement using Combination

$$^nP_r = \frac{n!}{(n-r)!}$$

$$^nC_r = \frac{n!}{(n-r)!\, r!}$$

^nP_r

$$\Rightarrow \quad ^nC_r = \frac{^nP_r}{r!} \qquad \Rightarrow \quad ^nP_r = {^nC_r} \cdot r!$$

$^nP_r = {}^nC_r \; r!$

Arrangement

Select r from n

Q] $\boxed{A, B, C, D}$ → Arrange in __ __

Sol^n : M1 $\underline{4}\ \underline{3} = 4 \times 3 = 12$

M2 $n \geq r \rightarrow {}^4P_2 = \dfrac{4!}{(4-2)!} = 12$

M3 ${}^4C_2 \times 2! = \dfrac{4 \times 3}{2 \times 1} \times 2 = 12$

(M, A, D). No of ways to arrange letters among themselves?

(M1)

MAD ADM DAM
MDA AMD DMA } 6 ways

(M2)

$\underline{3}\ \underline{2}\ \underline{1} = 3 \times 2 \times 1 = 6$

(M3)

$${}^{n}P_{r} = {}^{3}P_{3} = \frac{3!}{0!} = 3! = 6$$

A, A, B No of ways to arrange letters among themselves?

M1

AAB BAA

ABA

} 3 ways

M2

$$\frac{3!}{2!} = \frac{3 \times 2!}{2!} = 3$$

(Repeated letter का factorial)

9] SISTER. No of ways to arrange letters among themselves?

$$\frac{6!}{2!} = \frac{6\times5\times4\times3\times\cancel{2!}}{\cancel{2!}} = 360 \ //$$

9] ENGINEERING

Soln :

$$\frac{11!}{3!\,3!\,2!\,2!}$$

E N G I

(Vowels always come together)

~~E~~ D ~~U~~ C ~~A~~ T ~~I~~ ~~O~~ N

1. Vowels = a, e, i, o, u

2. D C T N $\boxed{E\ U\ A\ I\ O}$

$5! \times 5!$ //

(Vowels always come together)

PREPARATION

① P R P R T N $\boxed{E\ A\ A\ I\ O}$

$$\frac{7!\ 5!}{2!\ 2!\ 2!}$$

//

(Consonants always come together)

S I S T E R

I E $\boxed{S\ S\ T\ R}$

$$\frac{3!\ 4!}{2!}$$

//

(Vowels ~~always~~ never come together)

M $\cancel{A}$ C H $\cancel{I}$ N $\cancel{E}$

[Jo chahiye = Total − Jo NAHI chahiye]

Ans = $7! - 5! \times 3!$ //

(Vowels never ~~always~~ come together)

S ~~I~~ S T ~~E~~ R

① Total = $\frac{6!}{2!}$

② Jo NAHI chahiye = S S T R $\boxed{IE}$

$$\frac{5! \times 2!}{2!}$$

Ans = $\frac{6!}{2!} - \frac{5! \times 2!}{2!}$ //

(Vowels only occupy ODD position)

D E T A I L

_ _ _ _ _ _

1 2 3 4 5 6

$3! \times 3! = 36$

(Vowels only occupy ODD position)

M A C H I N E

$\underline{\quad}\ \overset{\checkmark}{\underline{\quad}}\ \underline{\quad}\ \overset{\checkmark}{\underline{\quad}}\ \underline{\quad}\ \overset{\checkmark}{\underline{\quad}}\ \underline{\quad}$

1 2 3 4 5 6 7

$4! \times 4!$ //

✓ ✓ ✓ ✓

$\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}$

1 3 5 7

$\{A, I, E\}$

$^4P_3$

$\Downarrow$

i) $\boxed{}$ __ __ __ $= 3!$

ii) __ $\boxed{}$ __ __ $= 3!$

iii) __ __ $\boxed{}$ __ $= 3!$

iv) __ __ __ $\boxed{}$ $= 3!$

$4 \times 3!$
$= 4!$

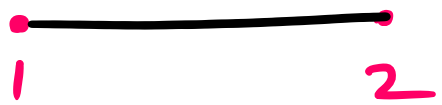
(Vowels only occupy ODD position)

P ~~O~~ ~~U~~ N D ~~I~~ N G

~~1~~ 2 ~~3~~ 4 ~~5~~ 6 ~~7~~ 8

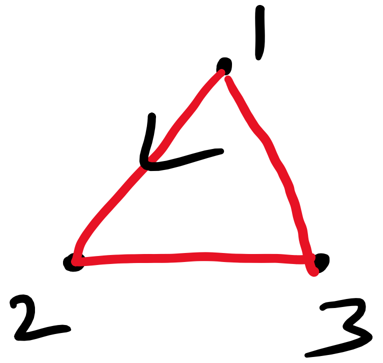
$$\frac{{}^4P_3 \times 5!}{2!} = \frac{4!\,5!}{2!}$$ //

Combination

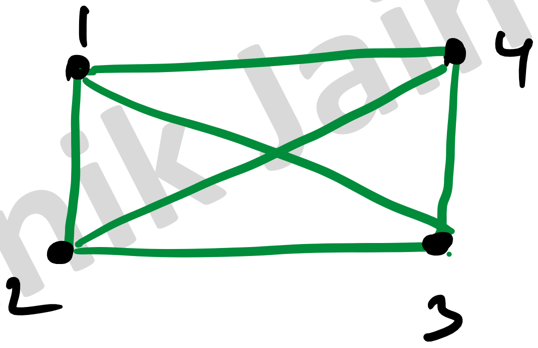


1 → 2

[2 point = 1 line]



1 → 2
2 → 3
3 → 1

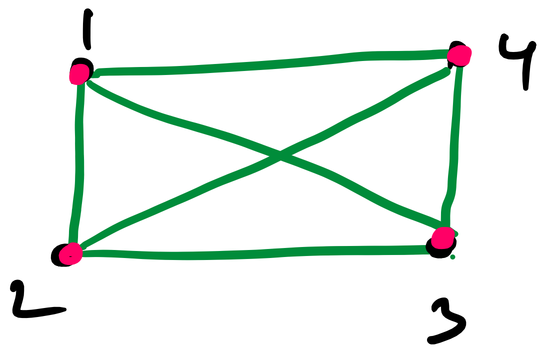


1 → 2
2 → 3
3 → 4
4 → 1

1 → 3
2 → 4

No of vertices = No of sides

No of Diagonals = $\frac{n}{2}(n-3)$



$\Rightarrow \quad {}^4C_2 \quad - \quad 4 \quad = 6-4 = 2$

$[D + S] \qquad [S]$ Diagonals

$\boxed{\text{n points}} \longrightarrow {}^nC_2 - n$

$[D+S] \qquad [S]$

$= \frac{n(n-1)}{2\times1} - n$

$= \frac{n^2 - n - 2n}{2}$

$= \frac{n^2-3n}{2} = \frac{n}{2}(n-3)$

Non collinear points

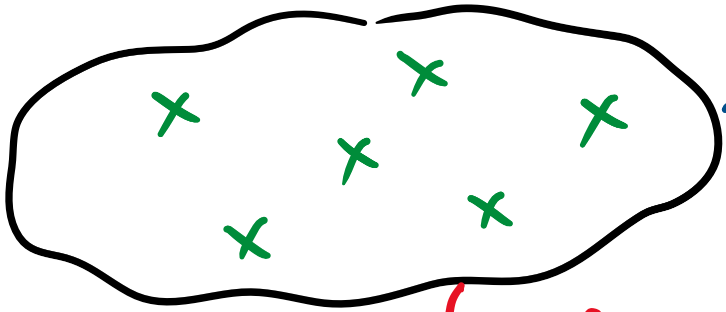
$${}^{10}C_2 = \frac{10 \times 9}{2 \times 1}$$

$$= 45$$

No of lines

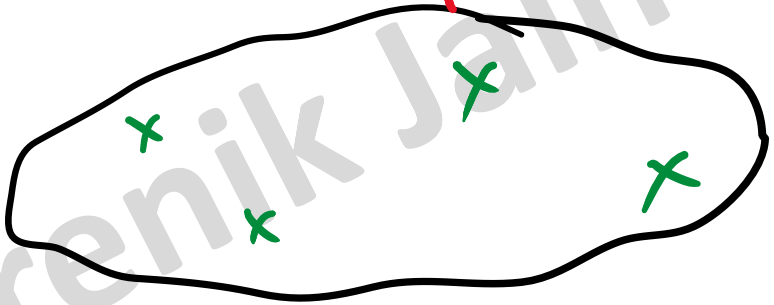
↳ ${}^{n}C_2$

Non collinear points



$$^6C_2 = \frac{6\times5}{2\times1} = 15$$

$$^6C_1 \times {}^4C_1 = 6\times4 = 24$$

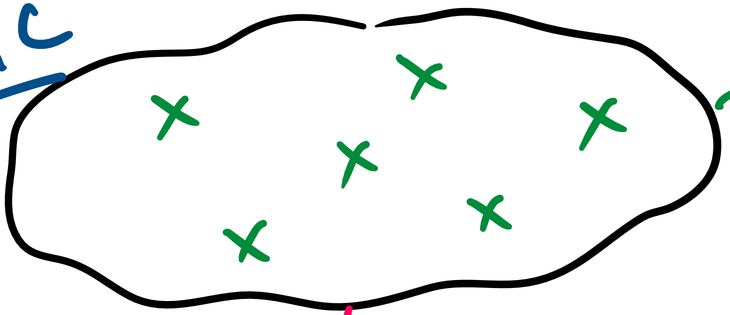


$$^4C_2 = \frac{4\times3}{2\times1} = 6$$

Ans $= 45$

collinear points

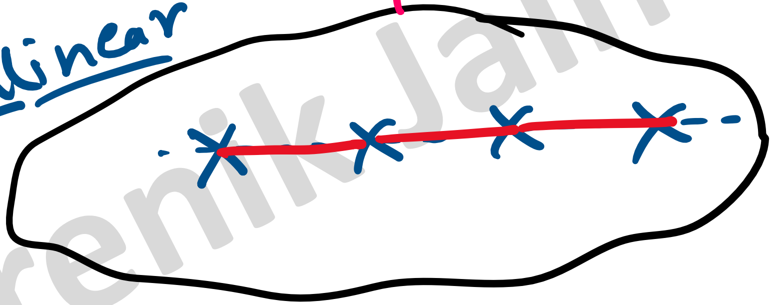
Non C



$^6C_2 = \frac{6\times5}{2\times1} = \underline{15}$

$^6C_1 \times {}^4C_1 = 6\times4 = \underline{24}$

Collinear



$\underline{1}$

$\boxed{\text{Ans} = 40}$

· formula

$${}^{n}C_2 - {}^{\text{No of collinear Pnts}}C_2 + 1$$

[Line + Wrong lines]

M2

Ans $= {}^{10}C_2 - {}^{4}C_2 + 1$

$= 40$

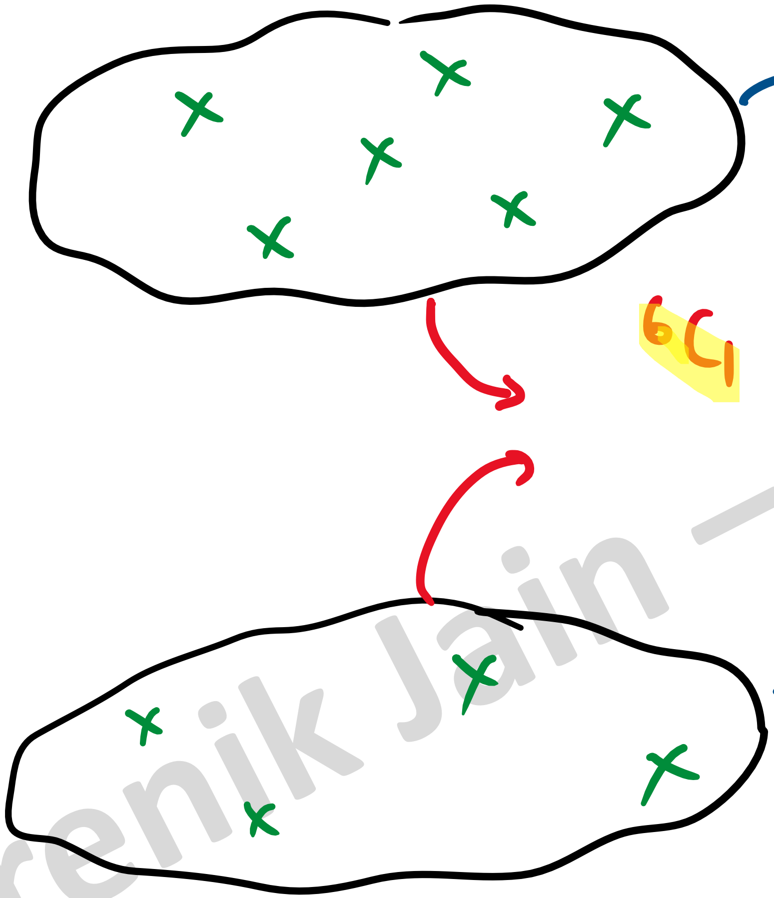
Triangle

[Non collinear pnts] → n points → no of Δ $= {}^{n}C_3$ //

$${}^{10}C_3 = \frac{10\times 9\times 8}{3\times 2\times 1}$$

$= 120$ //

Non collinear points



$$^6C_3 = \frac{6.5.4}{3.2.1} = 20$$

$$^6C_1 \times {}^4C_2 + {}^6C_2 \times {}^4C_1 = 96$$

$$\underbrace{^4C_3 = {}^4C_{4-3}} = {}^4C_1 = 4$$

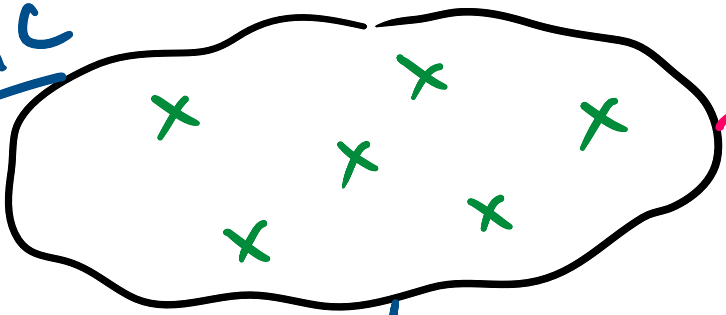
$$\left\{ {}^nC_r = {}^nC_{n-r} \right\}$$

Ans $= 120$

collinear points

Ans = 116

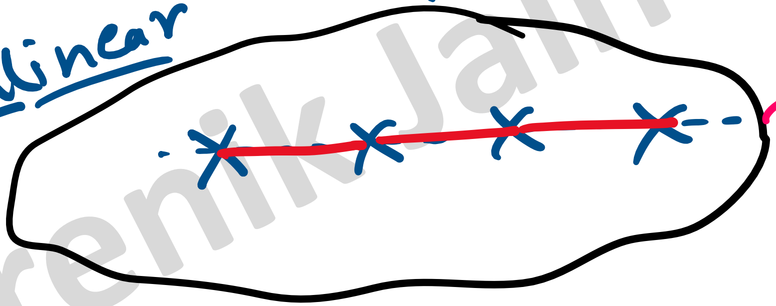
Non C



$^6C_3 = 20$

$^6C_2 \cdot {}^4C_1 + {}^6C_1 \cdot {}^4C_2 = 96$

Collinear



0

No of $\Delta = {}^nC_3 - {}^{\text{collinear pnts}}C_3$

[Δ + Not Δ]

M2

$$\text{Ans} = {}^{10}C_3 - {}^4C_3$$

$$= \frac{10\cdot 9\cdot 8}{3\cdot 2\cdot 1} - 4$$

$$= 116$$